

NAB2 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NAB2 gene fusions from the msk_impact_50k_2026 study, covering 72 fusion events. 25/72 fusions (34.7%) were in-frame. 49/72 fusions (68.1%) retained the NAB conserved region 2 (NCD2) domain. Domain retention was tested with Fisher's exact test ($p=0.11258$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for NAB2.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 72 analyzed fusion events, identifying 14 distinct fusion partner genes. The most frequent partner was STAT6, observed in 53 events.

Domain retention

NAB conserved region 2 (NCD2) domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 20/25 (80.0%) of in-frame fusions retained the domain, $p=0.11258$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Domain-disruption analysis was skipped: NAB2 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 71 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 372 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is 6 aa from the nearest configured domain boundary at 366 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: NAB2 has no group_field configured; confidence-stats analysis was skipped.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: NAB2 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
STAT6	53	0.736111111111112	0.20043213737826426	None	0.3434207302131831	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.460649253156967	1
CUX2	1	0.0138888888888888	0.20043213737826426	None	0.948051948051948	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2806892760271785	2
PDZRN4	1	0.0138888888888888	0.20043213737826426	None	0.948051948051948	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2806892760271785	3
MGAT4C	2	0.0277777777777777	0.20043213737826426	None	0.9090909090909091	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27829286291647964	4
KRT6A	1	0.0138888888888888	0.20043213737826426	None	0.935064935064935	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2779063446728186	5
LRP1	3	0.0416666666666666	0.20043213737826426	None	0.8658008658008658	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2749688060209942	6
TMEM117	2	0.0277777777777777	0.20043213737826426	None	0.6558441558441559	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22402570150646114	7
KIF5A	1	0.0138888888888888	0.20043213737826426	None	0.6623376623376623	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.21946478623126012	8
FMN1	3	0.0416666666666666	0.20043213737826426	None	0.23376623376623373	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.13953281344214447	9
DLGAP2	1	0.0138888888888888	0.20043213737826426	None	0.23376623376623373	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12762805153738258	10
FAM186A	1	0.0138888888888888	0.20043213737826426	None	0.23376623376623373	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12762805153738258	11
LFNG	1	0.0138888888888888	0.20043213737826426	None	None	None	['domain_retention', 'recurrence']	0.09868127456587769	12
LGR5	1	0.0138888888888888	0.20043213737826426	None	0.09740259740259738	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.09840727231660334	13
MTOR	1	0.0138888888888888	0.20043213737826426	None	0.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07753528715890391	14

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
319	0	49	3	19	0.0	0.02694427434170239	1.5695335082504807
320	0	49	4	18	0.0	0.00752854724253449	2.123288820003888
352	0	49	5	17	0.0	0.0020225947815764307	2.6940911176014084
364	0	49	13	9	0.0	8.560748975173139e-09	8.067488237477606
372	0	49	14	8	0.0	1.3283920823544527e-09	8.876673721601218
381	1	48	14	8	0.011904761904761904	1.731570731911155e-08	7.761559763513135
382	3	46	15	7	0.030434782608695653	1.0398205008988075e-07	6.983041624352472
400	4	45	15	7	0.04148148148148148	4.330878206429079e-07	6.36342402938942
426	4	45	16	6	0.03333333333333333	7.233726445978198e-08	7.140637918749475
434	5	44	16	6	0.04261363636363636	2.7039639334463545e-07	6.567999105485465
490	45	4	20	2	1.125	1.0	-0.0
502	46	3	20	2	1.5333333333333334	0.6419888956213136	0.19247248378799386
508	47	2	20	2	2.35	0.5828392348978783	0.23445122058408072
510	48	1	20	2	4.8	0.2249846907532149	0.6478470327668787
511	48	1	21	1	2.2857142857142856	0.5267605633802818	0.2783867465185951
521	49	0	21	1	None	0.3098591549295775	0.508835667896869

domain_retention tables

frame_domain_contingency_table

20	29
5	17

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	NAB conserved region 2 (NCD2)	71	49	13	9	0.35547087642896025

frequency tables

Partner_gene_counts

Partner_gene	Event_count
CUX2	1
DLGAP2	1
FAM186A	1
FMN1	3
KIF5A	1
KRT6A	1
LFNG	1
LGR5	1
LRP1	3
MGAT4C	2
MTOR	1
PDZRN4	1
STAT6	53
TMEM117	2

Per-event results (results.tsv)

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner_ gen e	fra me _sta tus	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost _do mai ns	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0052 994-T01-IM 6-1	P-0052 994-T0 1-IM6		FAM1 86A- NAB2	FAM 186 A	out- of-fr ame	thr ee_ _pr ime	574883 61	7	490	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2)		FAM186A (NM_001145475) - NAB2 (NM_005967) rearrangement: c.193-1572 6:FAM186A_c.1469-34:NA B2inv Protein Fusion: out of frame {FAM186A:NAB2}	NA
EVT-P-0015 989-T01-IM 6-2	P-0015 989-T0 1-IM6		FMN 1-NA B2	FMN 1	out- of-fr ame	thr ee_ _pr ime	574879 44	7	490	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2)		FMN1 (NM_001103184) - NAB2 (NM_005967) Rearrangement : t(12,15)(q 13.3,q13.3)(chr12:g.57487 944::chr15:g.33270869) Protein Fusion: out of frame {FMN1:NAB2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015 989-T05-IM 6-3	P-0015 989-T0 5-IM6		FMN 1-NA B2	FMN 1	out- of-fr ame	thr ee_ _pr ime	574879 44	7	490	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2)		FMN1 (NM_001103184) - NAB2 (NM_005967) rearrangement: t(12;15)(q1 3.3;q13.3)(chr12:g.574879 44::chr15:g.33270870) Protein Fusion: out of frame {FMN1:NAB2}	NA
EVT-P-0015 989-T02-IM 6-4	P-0015 989-T0 2-IM6		FMN 1-NA B2	FMN 1	out- of-fr ame	thr ee_ _pr ime	574879 45	7	490	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2)		FMN1 (NM_001103184) - NAB2 (NM_005967) Rearrangement : t(12,15)(q 13.3,q13.3)(chr12:g.57487 944::chr15:g.33270869) Protein Fusion: out of frame {FMN1:NAB2}	NA
EVT-P-0020 678-T01-IM 6-6	P-0020 678-T0 1-IM6		KRT6 A-NA B2	KRT 6A	in-fr ame	thr ee_ _pr ime	574868 17	5	382	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2)		KRT6A (NM_005554) - NAB2 (NM_005967) rearrangement: c.1204-574 :KRT6A_c.1144-29:NAB2in v Protein Fusion: in frame {KRT6A:NAB2}	NA
EVT-P-0045 531-T01-IM 6-7	P-0045 531-T0 1-IM6		LGR5 -NAB 2	LGR 5	unk now n	thr ee_ _pr ime	574884 58	7	511	False	lost	0.0	False		NAB cons erve d re gion 2 (N CD2)		LGR5 (NM_003667) - NAB2 (NM_005967) rearrangement: c.285-7692 :LGR5_c.1532:NAB2dup Protein Fusion: mid-exon {LGR5:NAB2}	NA
EVT-P-0068 029-T01-IM 7-9	P-0068 029-T0 1-IM7		LRP1 -NAB 2	LRP 1	unk now n	thr ee_ _pr ime	574871 82	6	426	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2)		LRP1 (NM_002332) - NAB2 (NM_005967) rearrangement: c.609:LRP 1_c.1277-8:NAB2dup Protein Fusion: mid-exon {LRP1:NAB2}	NA
EVT-P-0058 847-T01-IM 7-10	P-0058 847-T0 1-IM7		LRP1 -NAB 2	LRP 1	unk now n	thr ee_ _pr ime	574867 22	4	372	False	lost	0.0	False		NAB cons erve d re gion 2 (N CD2)		LRP1 (NM_002332) - NAB2 (NM_005967) rearrangement: c.5662+68 5:LRP1_c.1114:NAB2dup Protein Fusion: mid-exon {LRP1:NAB2}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gen_e	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0008 384-T01-IM 5-12	P-0008 384-T0 1-IM5		NAB2 -CUX 2	CUX 2	out- of-fr ame	thr_ee _pr ime	574864 06	3	364	True	disru pted	0.0179640 718562874 25	True			NAB co nserve d region 2 (NCD2)	CUX2 (NM_015267) - NAB2 (NM_005967) rearrangement: c.63+8305 8:CUX2_c.1091+42:NAB2 dup Protein fusion: out of frame (CUX2-NAB2)	NA
EVT-P-0038 472-T01-IM 6-13	P-0038 472-T0 1-IM6		NAB2 -DLG AP2	DLG AP2	out- of-fr ame	five _pr ime	574875 47	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - DLGAP2 (NM_004745) rearrangement: t(8;12)(p23 .3;q13.3)(chr8:g.1518542:: chr12:g.57487547) Protein Fusion: out of frame {NAB2:DLGAP2}	NA
EVT-P-0039 139-T01-IM 6-14	P-0039 139-T0 1-IM6		NAB2 -KIF5 A	KIF5 A	in-fr ame	five _pr ime	574860 74	3	320	True	disru pted	0.7245508 982035929	True			NAB co nserve d region 2 (NCD2)	NAB2 (NM_005967) - KIF5A (NM_004984) rearrangement: c.958-157: NAB2_c.129+5819:KIF5Ad el Protein Fusion: in frame {NAB2:KIF5A}	NA
EVT-P-0022 192-T01-IM 6-18	P-0022 192-T0 1-IM6		NAB2 -LRP 1	LRP 1	out- of-fr ame	thr_ee _pr ime	574865 54	4	364	True	disru pted	0.0179640 718562874 25	True			NAB co nserve d region 2 (NCD2)	LRP1 (NM_002332) - NAB2 (NM_005967) rearrangement: c.842-2141 :LRP1_c.1092-146:NAB2d up Protein Fusion: out of frame {LRP1:NAB2}	NA
EVT-P-0028 347-T01-IM 6-21	P-0028 347-T0 1-IM6		NAB2 -MGA T4C	MG AT4 C	unk now n	five _pr ime	574863 29	3	352	False	disru pted	0.9161676 646706587	True			NAB co nserve d region 2 (NCD2)	NAB2 (NM_005967) rearrangement: c.1056:NA B2_chr12:g.86871854inv Protein Fusion: mid-exon {NAB2:MGAT4C}	NA
EVT-P-0011 161-T01-IM 5-22	P-0011 161-T0 1-IM5		NAB2 -MGA T4C	MG AT4 C	unk now n	thr_ee _pr ime	574865 34	4	364	True	disru pted	0.0179640 718562874 25	True			NAB co nserve d region 2 (NCD2)	NAB2 (NM_005967) rearrangement: c.1092-166 :NAB2_chr12:g.87188994d el Antisense fusion	NA
EVT-P-0025 245-T01-IM 6-23	P-0025 245-T0 1-IM6		NAB2 -MTO R	MT OR	unk now n	thr_ee _pr ime	574919 19	7	526	True	lost	0.0	False		NAB cons erve d re gion 2 (N CD2)		MTOR (NM_004958) - STAT6 (NM_003153) rearrangement: t(1;12)(p36 .22;q13.3)(chr1:g.1130031 0::chr12:g.57491919) Transcript Fusion {MTOR:NAB2}	NA
EVT-P-0001 055-T02-IM 5-49	P-0001 055-T0 2-IM5		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574879 38	7	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exon 6 fused with STAT6 exon16) : c.14 69-457:NAB2_c.1760:STA T6inv Protein fusion: mid-exon (NAB2-STAT6)	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0066 534-T01-IM 7-50	P-0066 534-T0 1-IM7		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574880 34	7	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-361 :NAB2_c.1775:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0003 845-T01-IM 5-51	P-0003 845-T0 1-IM5		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574878 34	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) -STAT6 (NM_003153) fusion (NAB2 exon 6 fused to STAT6 exon 16) : c.146 8+453:NAB2_c.1744+30:S TAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0029 101-T01-IM 6-52	P-0029 101-T0 1-IM6		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574868 24	5	382	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1144-22:NAB2_c. -1269:STAT6inv Transcript Fusion {NAB2:STAT6}	NA
EVT-P-0029 763-T01-IM 6-53	P-0029 763-T0 1-IM6		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574875 87	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1468+206:NAB2_ c.1624:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0001 055-T03-IM 6-55	P-0001 055-T0 3-IM6		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574879 38	7	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-457 :NAB2_c.1760:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0020 523-T01-IM 6-56	P-0020 523-T0 1-IM6		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574875 05	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused in frame with STAT6 exons 16-22): c.1468+124: NAB2_c.1745-112:STAT6i nv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0032 069-T01-IM 6-57	P-0032 069-T0 1-IM6		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574877 12	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1468+331:NAB2_ c.1870:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022 259-T02-IM 6-61	P-0022 259-T0 2-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57488455	7	510	False	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-7 fused with STAT6 exons 16-22): c.1529:NAB2_c.1745-128:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0004 282-T03-IM 5-62	P-0004 282-T0 3-IM5		NAB2-STAT6	STAT6	in-frame	five_prime	57488115	7	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exon 16 to 22) : c.1469-280:NAB2_c.1745-157:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0033 539-T02-IM 6-63	P-0033 539-T0 2-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57486902	5	400	False	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) rearrangement: c.1200:NA B2_chr12:g.57486902inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0019 357-T03-IM 6-64	P-0019 357-T0 3-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57488027	7	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1469-368:NAB2_c.1690:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0019 182-T01-IM 6-66	P-0019 182-T0 1-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57487665	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused in frame with STAT6 exons 16-22): c.1468+284: NAB2_c.1744+39:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0036 288-T01-IM 6-67	P-0036 288-T0 1-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57488128	7	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion: c.1469-267:NAB2_c.1850:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0036 975-T01-IM 6-68	P-0036 975-T0 1-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57486781	4	381	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) rearrangement: c.1143+30: NAB2_chr12:g.57508083inv Transcript Fusion {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019 032-T01-IM 6-69	P-0019 032-T0 1-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57488347	7	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 15-22): c.1469-48:NAB2_c.1691:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0017 518-T01-IM 6-70	P-0017 518-T0 1-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57486597	4	364	True	disrupted	0.9880239520958084	True			NAB conserved region 2 (NCD2)	NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-2 fused to STAT6 exons 18-22): c.1092-103:NAB2_c.2042:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0016 751-T02-IM 6-71	P-0016 751-T0 2-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57487958	7	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1469-437:NAB2_c.1745-109:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0038 591-T01-IM 6-72	P-0038 591-T0 1-IM6		NAB2-STAT6	STAT6	in-frame	five_prime	57487747	6	490	True	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exon6 fused with STAT6 exon16) : c.1468+366:NAB2_c.1745-67:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0016 113-T01-IM 6-73	P-0016 113-T0 1-IM6		NAB2-STAT6	STAT6	unknown	five_prime	57486482	3	364	True	disrupted	0.9880239520958084	True			NAB conserved region 2 (NCD2)	NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exon3 fused to STAT6 exon 18: c.1091+118:NAB2_c.2051:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0004 405-T01-IM 5-74	P-0004 405-T0 1-IM5		NAB2-STAT6	STAT6	unknown	five_prime	57487215	6	434	False	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) -STAT6(NM_003153) Fusion (NAB2 exon 6 fused to STAT6 exon 18) : c.1302:NAB2_c.1969:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0004 943-T01-IM 5-75	P-0004 943-T0 1-IM5		NAB2-STAT6	STAT6	unknown	five_prime	57488487	7	521	False	retained	1.0	False	NAB conserved region 2 (NCD2)			NAB2 (NM_005967) - STAT6 (NM_003153) reciprocal fusion (NAB2 exons 1-7 with STAT6 exons 15-22) : c.1561:NAB2_c.1739:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040 364-T01-IM 6-76	P-0040 364-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57487738	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+357:NAB2_c.1892-42:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0040 364-T02-IM 6-77	P-0040 364-T0 2-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57487738	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+357:NAB2_c.1892-42:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0014 386-T01-IM 6-78	P-0014 386-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57487858	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 17 - 22): c.1468+477:NAB2_c.1891+67:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0043 579-T01-IM 6-79	P-0043 579-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57488047	7	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-348:NAB2_c.1744+96:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0044 036-T01-IM 6-80	P-0044 036-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57488037	7	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-358:NAB2_c.1744+93:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0044 232-T02-IM 6-81	P-0044 232-T0 2-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57488269	7	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-126:NAB2_c.1700:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0001 947-T03-IM 6-82	P-0001 947-T0 3-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57487383	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+2:NAB2_c.1769:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0045 590-T01-IM 6-83	P-0045 590-T0 1-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57487780	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+399:NAB2_c.1789:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0046 398-T01-IM 6-84	P-0046 398-T0 1-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57487473	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92: NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0046 398-T02-IM 6-85	P-0046 398-T0 2-IM6		NAB2- STAT6	STAT6	unknown	five_prime	57487473	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+92: NAB2_c.1705:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0048 656-T01-IM 6-86	P-0048 656-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57486417	3	364	True	disrupted	0.9880239520958084	True			NAB conserved region 2 (NCD2)	NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1091+53: NAB2_c.1956-25:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0013 489-T01-IM 5-87	P-0013 489-T0 1-IM5		NAB2- STAT6	STAT6	unknown	five_prime	57487506	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6(NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exons 15 to 22) : c.1468+125:NAB2_c.1625:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0053 200-T02-IM 6-89	P-0053 200-T0 2-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57487609	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+228:NAB2_c.1891+45:STAT6 inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0013 261-T01-IM 5-90	P-0013 261-T0 1-IM5		NAB2- STAT6	STAT6	unknown	five_prime	57487567	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exon 6 fused to STAT6 exon 15): c.1468+186:NAB2_c.1703:STAT6 inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0013 216-T02-IM 5-91	P-0013 216-T0 2-IM5		NAB2- STAT6	STAT6	in-frame	five_prime	57487806	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion: c.1468+425:NAB2_c.1745-142:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0055 865-T01-IM 6-92	P-0055 865-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57488325	7	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-70: NAB2_c.1744+36:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA

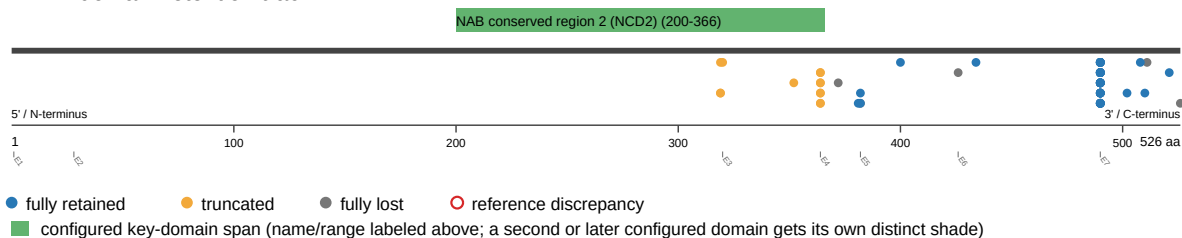
event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_ooint_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0055 909-T01-IM 6-93	P-0055 909-T0 1-IM6		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574859 66	2	319	True	disru pted	0.7185628 74251497	True			NAB co nserve d region 2 (NCD2)	NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.957+185: NAB2_c.340-42:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0012 841-T01-IM 5-94	P-0012 841-T0 1-IM5		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574876 86	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 with STAT6 exons 15-22) : c.1468+305:NAB2_c.1712: STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0057 770-T01-IM 6-95	P-0057 770-T0 1-IM6		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574874 67	6	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+86: NAB2_c.1744+3:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0057 786-T01-IM 6-96	P-0057 786-T0 1-IM6		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574882 44	7	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-151 :NAB2_c.1892-23:STAT6in v Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0012 421-T01-IM 5-97	P-0012 421-T0 1-IM5		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574865 09	3	364	True	disru pted	0.9880239 520958084	True			NAB co nserve d region 2 (NCD2)	NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-3 fused in-frame with STAT6 exons 20-22): c.1091+145: NAB2_c.2159+239:STAT6i nv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0007 527-T01-IM 5-98	P-0007 527-T0 1-IM5		NAB2 -STA T6	STA T6	in-fr ame	five _pr ime	574881 11	7	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) reciprocal fusion (NAB2 exons 1-6 with STAT6 exons 16-22) : c.1469-284: NAB2_c.1745-161:STAT6i nv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0001 055-T04-IM 6-99	P-0001 055-T0 4-IM6		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574879 38	7	490	True	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-457 :NAB2_c.1760:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0059 117-T01-IM 7-100	P-0059 117-T0 1-IM7		NAB2 -STA T6	STA T6	unk now n	five _pr ime	574884 48	7	508	False	retai ned	1.0	False	NAB c onserv ed region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1522:NA B2_c.287:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059 275-T01-IM 7-101	P-0059 275-T0 1-IM7		NAB2- STAT6	STAT6	unknown	five_prime	57487481	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+100:NAB2_c.1801:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0006 897-T01-IM 5-102	P-0006 897-T0 1-IM5		NAB2- STAT6	STAT6	unknown	five_prime	57488251	7	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2(NM_005967) - STAT6(NM_003153) Fusion (NAB2 exons 1 to 6 fused with STAT6 exons 16 to 22) : c.1469-144:NAB2_c.1811:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0006 243-T01-IM 5-103	P-0006 243-T0 1-IM5		NAB2- STAT6	STAT6	unknown	five_prime	57488431	7	502	False	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2(NM_005967) -STAT6(NM_003153) Fusion (NAB2 exons 1 to 7 fused with STAT6 exon 16 to 22) : c.1505:NAB2_c.1745-142:STAT6inv Protein fusion: mid-exon (NAB2-STAT6)	NA
EVT-P-0005 578-T01-IM 5-104	P-0005 578-T0 1-IM5		NAB2- STAT6	STAT6	in-frame	five_prime	57486815	5	382	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) Fusion (NAB2 exon4 fused with STAT6 exon5) : c.1144-31:NAB2_c.340-81:STAT6inv Protein fusion: in frame (NAB2-STAT6)	NA
EVT-P-0067 841-T01-IM 7-106	P-0067 841-T0 1-IM7		NAB2- STAT6	STAT6	unknown	five_prime	57487573	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1468+192:NAB2_c.1861:STAT6inv Protein Fusion: mid-exon {NAB2:STAT6}	NA
EVT-P-0066 305-T01-IM 7-107	P-0066 305-T0 1-IM7		NAB2- STAT6	STAT6	in-frame	five_prime	57488138	7	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) rearrangement: c.1469-257:NAB2_c.1745-14:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA
EVT-P-0027 038-T01-IM 6-108	P-0027 038-T0 1-IM6		NAB2- STAT6	STAT6	in-frame	five_prime	57487452	6	490	True	retained	1.0	False	NAB conserved region 2 (NC D2)			NAB2 (NM_005967) - STAT6 (NM_003153) fusion (NAB2 exons 1-6 fused to STAT6 exons 16-22): c.1468+71:NAB2_c.1744+14:STAT6inv Protein Fusion: in frame {NAB2:STAT6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030 371-T02-IM 6-109	P-0030 371-T0 2-IM6		NAB2 -TME M117	TME M11 7	out- of-fr ame	five _pr ime	574858 85	2	319	True	disru pted	0.7185628 74251497	True			NAB co nserve d region 2 (NCD2)	NAB2 (NM_005967) - TMEM117 (NM_032256) rearrangement: c.957+104: NAB2_c.278-424:TMEM11 7dup Protein Fusion: out of frame {NAB2:TMEM117}	NA
EVT-P-0004 638-T01-IM 5-110	P-0004 638-T0 1-IM5		PDZ RN4- NAB2	PDZ RN4	out- of-fr ame	thr ee _pr ime	574866 31	4	364	True	disru pted	0.0179640 718562874 25	True			NAB co nserve d region 2 (NCD2)	PDZRN4 (NM_001164595) - NAB2 (NM_005967) - rearrangement: c.843+122 859:PDZRN4_c.1092-69:N AB2del. Protein fusion: out of frame (PDZRN4-NAB2)	NA
EVT-P-0030 371-T01-IM 6-111	P-0030 371-T0 1-IM6		TME M117 -NAB 2	TME M11 7	out- of-fr ame	five _pr ime	574858 85	2	319	True	disru pted	0.7185628 74251497	True			NAB co nserve d region 2 (NCD2)	NAB2 (NM_005967) - TMEM117 (NM_032256) Rearrangement : c.957+10 4:NAB2_c.278-424:TMEM 117dup Protein Fusion: out of frame {NAB2:TMEM117}	NA

domain retention outliers

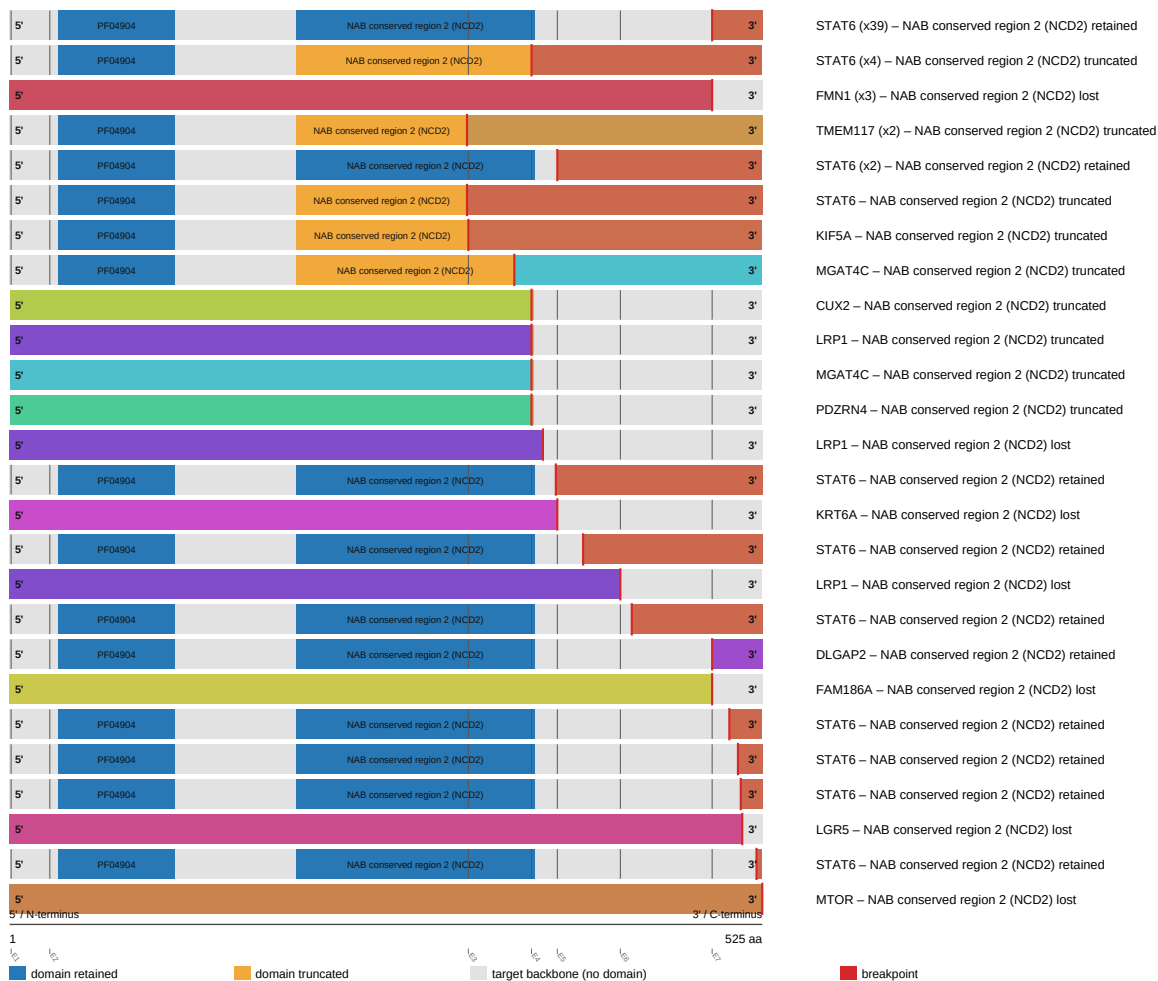
NAB2 domain-retention track



fusion schematic

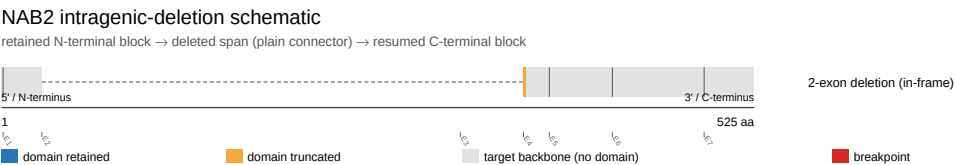
NAB2 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (NAB2 portion is domain-colored, exon ticks)



Partner-block colors are decorative: row labels identify partners.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

